SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 5

Prescriptive Analytics: Optimisation

Prescriptive analytics recommends the best action within limits and prices each choice: decision models, LP (Region 51 doctors vs beds), shadow prices, scaling (400 doctors, price of fairness), integers (nurse roster), diminishing returns (GRG), uncertainty (platelets, critical ratio) and six traps.

Material

1 Purpose and anatomy of a decision model

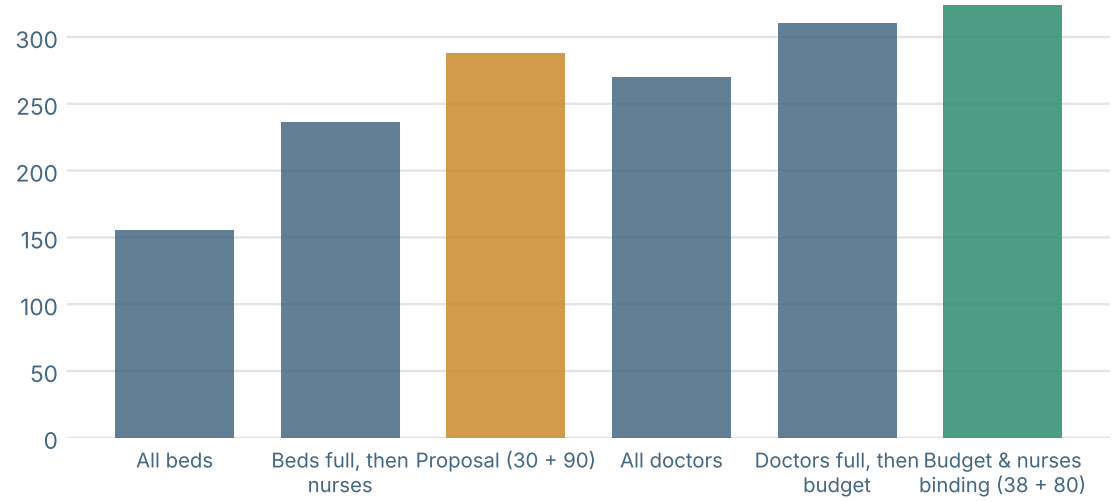
Purposes: choose the best, use scarce resources, expose trade-offs, decide at scale, plan under uncertainty. Kidney exchange (Roth & Shapley, Nobel 2012). Models have decisions, one objective and numeric constraints.

2 Linear programming: Region 51

Region 51 LP: max 6D + 1.2B subject to budget, nurses and room. The proposal scores 288; the optimal corner D=38, B=80 scores 324 (+12.5%). Beds win per rupiah, doctors per nurse, so the best plan mixes both.

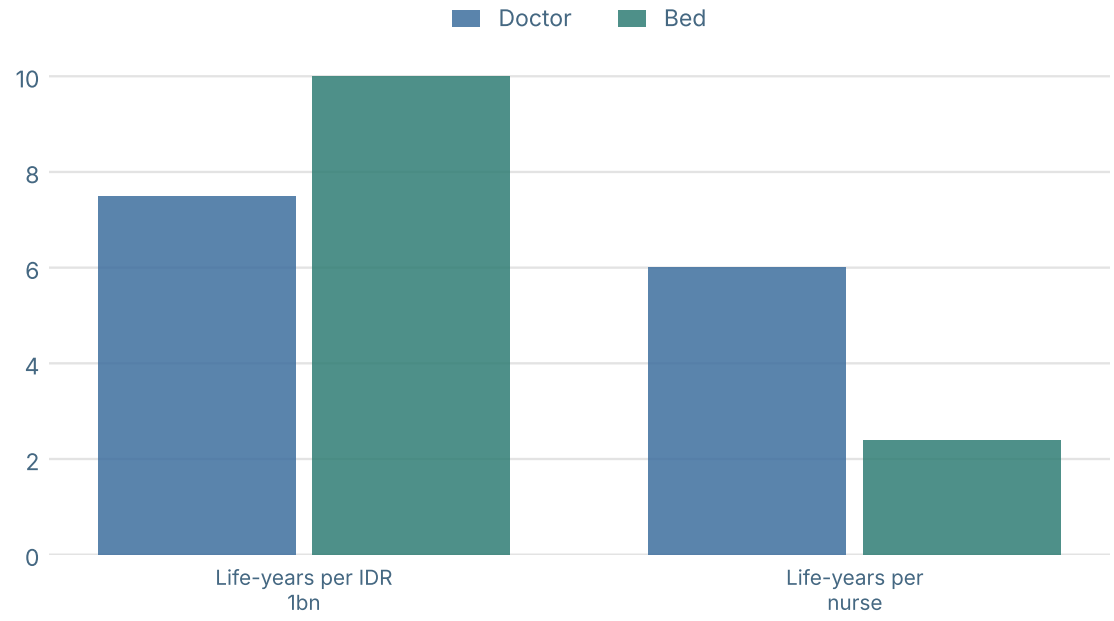
3 Region 51: corner points and simulator Deep dive

Life-years at each corner



The proposal (288) is inside the region; the optimum is the budget-nurse corner (324).

Why a mix: which resource each wins

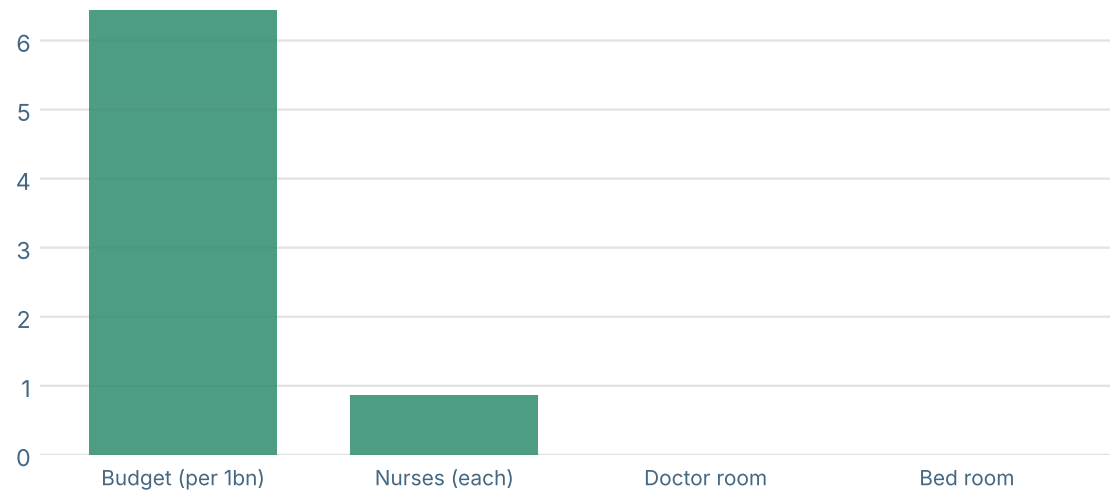


4 Shadow prices

Budget shadow price 6.43 life-years per IDR bn, nurse 0.86, room 0. An agency nurse at IDR 120m costs 0.77 → worth it; at 150m (0.96) not. Extra IDR 2bn ≈ +12.9 life-years.

5 Shadow prices in a chart Deep dive

Shadow price per constraint



Non-binding constraints are worth 0.

Agency nurse 120m
0.77 < 0.86 → worth it.

150m
0.96 > 0.86 → not worth it.

6 Scale, integers, diminishing returns

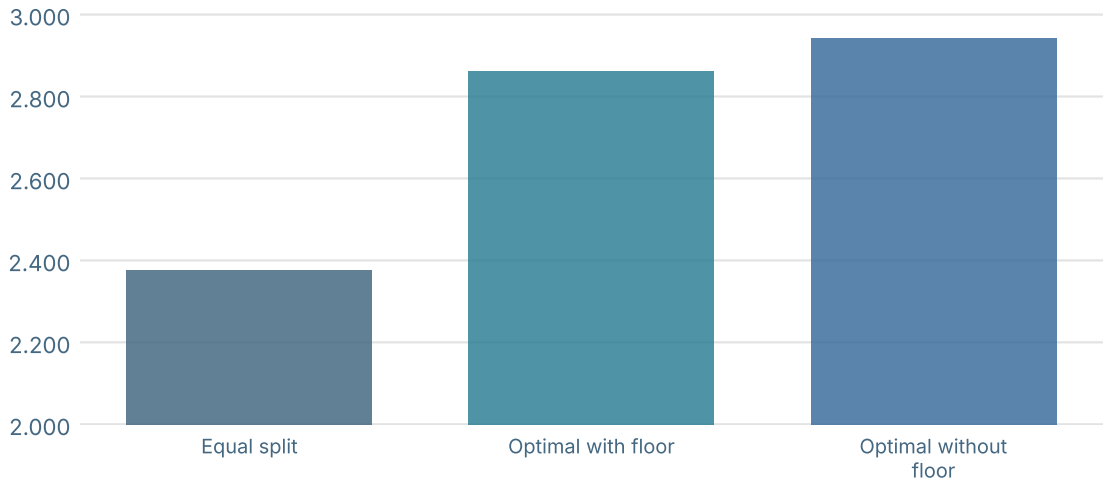
400 doctors: optimal 2,862 vs equal split 2,376 (+20%); the fairness floor costs 80 life-years; beware Region 29's imputed blank. Nurse roster: fractions 21.4, rounding 27, integer 22 (save ~IDR 390m/yr). Diminishing returns: GRG gives 716

vs 679 equal and 635 straight-line; marginal values equalise (~4.6). Match Solver method to model type.

7 Scale, integers and diminishing returns

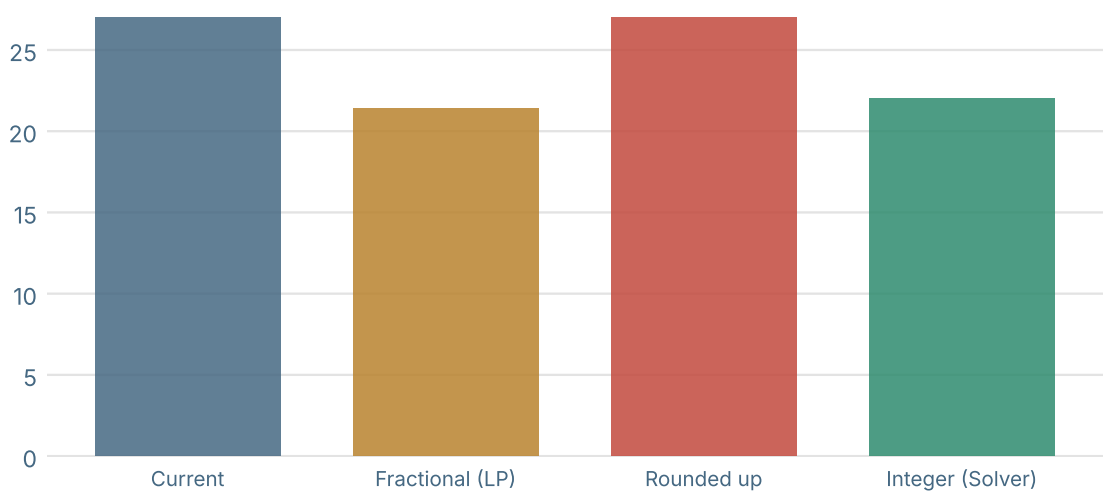
Deep dive

400 doctors for 53 regions



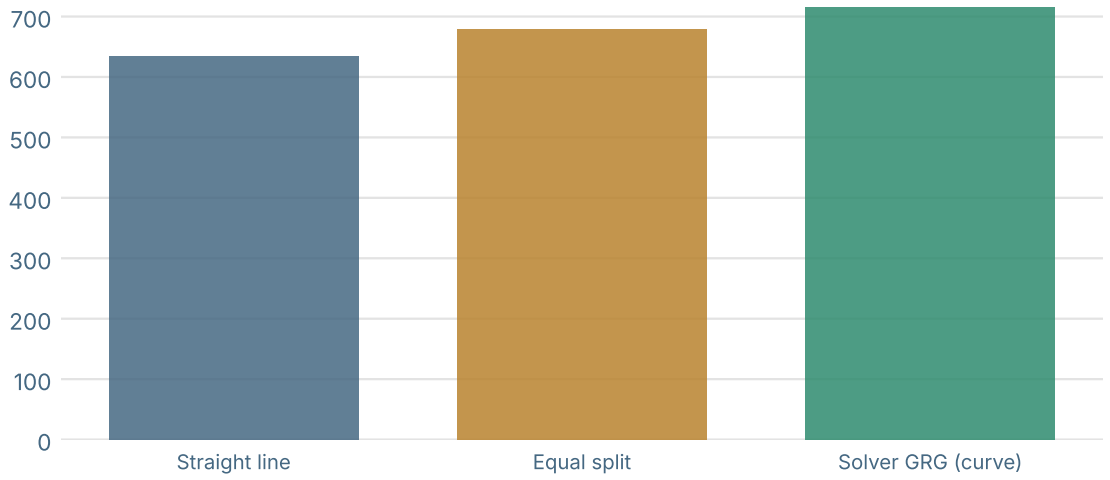
The fairness floor costs 80 life-years a year.

ER nurse roster



Rounding 21.4 up to 27 kills the savings; an integer model gives 22.

Diminishing returns: life-years



8 Uncertainty and six traps

Platelets: lowest cost at 122 units (not the 110 average); critical ratio 2.5/3.1 = 0.81; with IDR 1.5m waste cost, 0.63 → ~114. Monte Carlo confirms. Six traps: wrong objective, missing constraint, bad inputs, wrong (correlational) model, false precision, nobody follows it.

9 Platelet stock: asymmetric costs Deep dive




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CR = 2.5/3.1 = 0.81 → ~122 units
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Key takeaways

1. Data shows the pain; decisions need objective and constraints.
2. LP optimum at a corner.
3. Shadow prices.
4. Integer constraints, don't round.
5. The benefit curve shapes the plan.
6. Weigh the two errors.

Case studies

Case & worked answers

Answers to the Excel exercises

Exercise answers: 324 life-years (shadow prices 6.43, 0.86); 2,862 vs 2,376; nurses 21.4 → 27 → 22; GRG 716; platelets 122 (114). Solver setup: objective, changing cells, Used ≤ Limit, non-negative, Simplex LP, SUMPRODUCT.

Worked examples

Problem 1 Medium **Program linear dengan titik sudut**

Maksimalkan laba 5x + 4y dengan batasan: 6x + 4y ≤ 24 (jam mesin), x + 2y ≤ 6 (bahan), x, y ≥ 0.

1. Titik sudut di sumbu: (0,0) → 0; (4,0) → 20; (0,3) → 12.
2. Perpotongan dua batas: dari x + 2y = 6 → x = 6 − 2y. Masukkan: 6(6 − 2y) + 4y = 24 → 36 − 8y = 24 → y = 1,5; x = 3.
3. Laba di (3; 1,5) = 15 + 6 = 21.
4. Cek batas: 6×3 + 4×1,5 = 24 ✓; 3 + 2×1,5 = 6 ✓ → keduanya binding.

Answer: Optimal x = 3, y = 1,5, laba 21.

Problem 2 Medium **Critical ratio (stok koran)**

Kekurangan 1 unit = laba hilang Rp40.000. Kelebihan 1 unit = rugi Rp15.000. Permintaan normal, rata-rata 200, standar deviasi 30. Berapa stok optimal?

1. `CR = Cu ÷ (Cu + Co) = 40.000 ÷ 55.000 = 0,727`
2. Cari z dengan P(Z ≤ z) = 0,727 → z ≈ 0,60.
3. Stok = 200 + 0,60 × 30 = 218.

Answer: ± 218 unit — di atas rata-rata, karena kekurangan lebih mahal dari kelebihan.

Problem 3 Easy **Memakai shadow price**

Shadow price jam mesin = Rp2.500 per jam (berlaku untuk tambahan hingga 10 jam). Vendor menawarkan sewa mesin tambahan Rp2.000/jam. Ambil?

1. Setiap jam tambahan menaikkan laba Rp2.500 dan biayanya Rp2.000.
2. Untung bersih Rp500 per jam, maksimal 10 jam (rentang berlaku) → Rp5.000.

Answer: Ambil, maksimal 10 jam. Di atas 10 jam, jalankan ulang model karena shadow price bisa berubah.

Practice questions

Try answering first. The answer key is below each question.

1. Region 51's optimum is...

A. 30 + 90
B. 38 + 80
C. 45 + 0
D. 0 + 129

Answer: B. 324 life-years.

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2. Budget shadow price 6.43 means...

A. Cost per doctor
B. +1bn adds ~6.43 life-years (within range)
C. Total budget
D. Per bed

Answer: B. Within the allowable range.

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3. 'Enough staff' is...

A. A constraint
B. A wish — make it a number
C. An objective
D. A decision

Answer: B. ≥4 nurses per shift is.

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4. Cu 2.5, Co 0.6. Critical ratio...

A. 0.19
B. 0.63
C. 0.81
D. 4.17

Answer: C. 2.5/3.1.

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5. Solver sends 82 doctors but only 30 applied. Pitfall...

A. Wrong objective
B. Missing constraint
C. False precision
D. Bad input

Answer: B. Add the applicant limit.

Sources

- W5_Prescriptive_Analytics (MK6)
- Health Exercise data (53 region)
- Nobel Ekonomi 2012 — Roth & Shapley